ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PAPER 1 · PRACTICE B

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 8 year: roof trusses, types of structures, arch bridges, loads, strengthening methods and design specifications, plastics, composites and vacuum forming, gear and belt calculations, rack and pinion, chain drives, OR logic circuits and electrical safety.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings and tables.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A bridge or beam that is supported at one end only is called a …

A cantilever. B truss. C suspension bridge. D arch.

1.2 A member of a structure that is in tension is called a …

A strut. B tie. C column. D pillar.

1.3 A worm and wheel is used to give …

A a large increase in speed. B linear motion only.
C no change in speed. D a very large reduction in speed.

1.4 A driver gear with 60 teeth turns a driven gear with 20 teeth. The driven gear turns …

A three times faster than the driver. B three times slower than the driver.
C at the same speed. D six times faster.

1.5 Gears that change the direction of rotation through 90° are …

A spur gears. B idler gears. C bevel gears. D racks.

1.6 Two switches connected in parallel to control one lamp work like …

A an AND gate. B a NOT gate. C an OR gate. D a resistor.

1.7 A belt that keeps slipping on its pulleys can be fixed by …

A tightening the belt. B oiling the belt.
C making the pulleys smaller. D crossing the belt.

1.8 Which plastic cannot be softened and reshaped by heating?

A Thermoplastic B Polythene
C Acrylic D Thermosetting plastic

1.9 In first-angle orthographic projection, the left view is drawn …

A to the left of the front view. B to the right of the front view.
C above the front view. D below the top view.

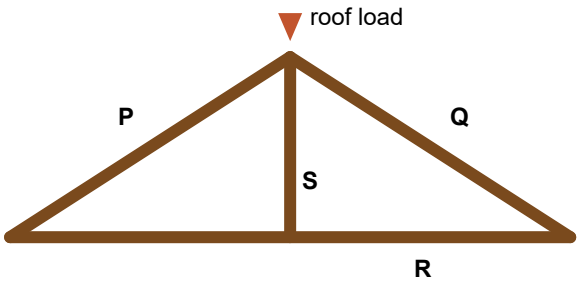
1.10 An electronic component that allows current to flow in one direction only is a …

A resistor. B switch. C buzzer. D diode.

[10]

QUESTION 2: STRUCTURES

2.1 Study the simple roof truss.



2.1.1 Member R is pulled outwards by the rafters. Is R a strut or a tie? (1)

2.1.2 Why are roof trusses made in the shape of triangles? (2)

2.2 Give ONE example of each: (a) a frame structure; (b) a shell structure; (c) a solid structure. (3)

2.3 Explain how an arch bridge carries the weight of a heavy truck. (2)

2.4 Explain the difference between a static load and a dynamic load on a bridge. (2)

2.5 Explain how each method makes a material or structure stronger: (a) corrugating (folding) a sheet; (b) using a tube instead of a solid rod; (c) adding a gusset to a joint. (3)

2.6 Your class must build a model bridge from drinking straws. Write TWO specifications and ONE constraint for the bridge. (3)

2.7 Why do engineers build and test models before they build a real bridge? (2)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 (a) Give TWO advantages of plastic as a material. (b) Explain ONE environmental problem caused by plastic. (4)

3.2 Name the plastic that has (a) recycling code 1 and (b) recycling code 2. (2)

3.3 What is a composite material? Give ONE example. (2)

3.4 Describe the process of vacuum forming in THREE steps. (3)

3.5 Why are the handles of screwdrivers usually made of plastic? (1)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 A motor turns a driver gear with 12 teeth at 1 500 r/min. It meshes with a driven gear with 36 teeth. (a) Write down the velocity ratio. (b) Calculate the speed of the driven gear. (c) Is the turning force (torque) at the driven gear bigger or smaller than at the driver? (4)

4.2 In a washing machine, a motor pulley with a diameter of 50 mm turns at 1 400 r/min. It drives the drum pulley, which has a diameter of 350 mm. (a) Calculate the speed of the drum. (b) Write down the velocity ratio. (3)

4.3 A pinion with 20 teeth turns one full revolution. The teeth of the rack are 5 mm apart. How far does the rack move? (2)

4.4 Give TWO advantages of a chain drive compared with a belt drive. (2)

4.5 Draw a systems diagram (input → process → output) for a sliding gate that is opened by an electric motor and a rack and pinion. (3)

4.6 A single fixed pulley has a mechanical advantage of 1. Why is it still useful, for example on a flagpole? (2)

4.7 Why should gears and chains be lubricated? (2)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 The interior light of a car must switch on when the driver's door OR the passenger's door is opened. Each door has its own switch. Draw a neat circuit diagram for this system using the correct symbols. (4)

5.2 Draw and complete the truth table for this system. Use 0 for closed door/off and 1 for open door/on. (4)

5.3 (a) What is the function of a resistor in a circuit? (b) What happens to the brightness of a lamp if a larger resistor is added in series with it? (2)

5.4 Name TWO household appliances that contain an electric motor. (2)

5.5 What is earthing, and why is it important for appliances with metal casings? (2)

5.6 What is a short circuit, and why is it dangerous? (2)

5.7 A burglar alarm must ring if a door OR a window is opened. Which type of logic is used? (1)

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TOTAL: 75

END OF QUESTION PAPER